DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Count Data Models: Poisson Regression

Learning Objectives

- Describe the Poisson distribution
- Develop a Poisson regression model
- Estimate a Poisson regression model
- Interpret Poisson model coefficients
- Examine potential overdispersion

Count Data

- Limited Dependent Variable Models:
  - Binary Response
  - Corner Solution Response
  - Censored Model
  - Truncated Model
  - Count Data

- Count Data:
  - Dependent variable takes non-negative integer values
    - Especially when those numbers are relatively small
    - E.g., number of children in a family; number of store visits every month; number of posts written on social media…
  - Simple linear models cannot approximate well

Poisson Distribution

- The Poisson Distribution:

$$P(y = k|\lambda) = \frac{\lambda^k e^{-\lambda}}{k!},\ k = 0,1,\ldots\ \ (\lambda > 0)$$

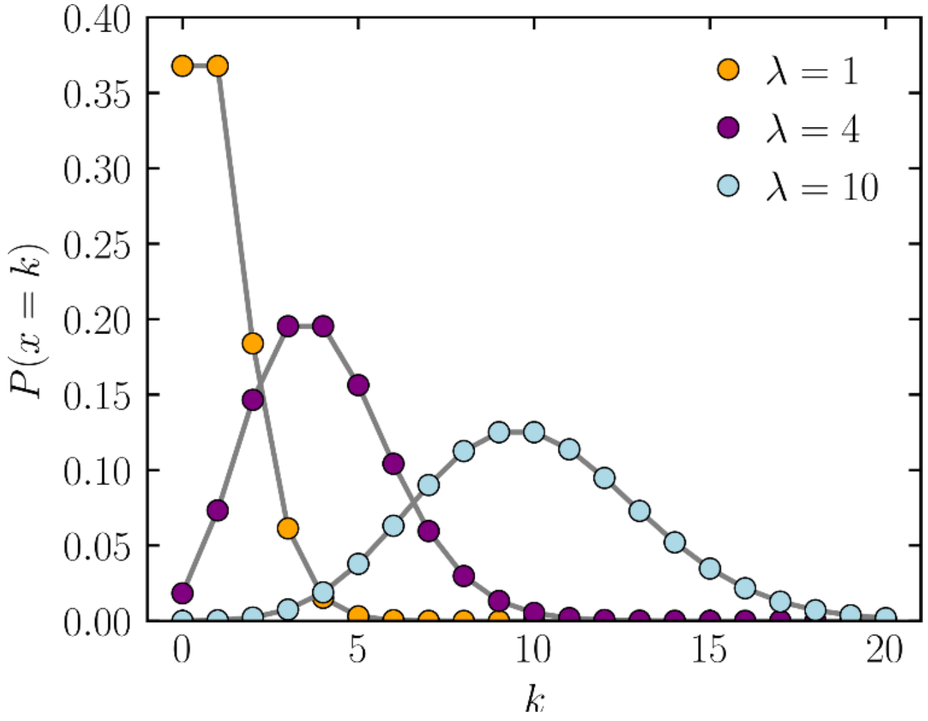
  - Discrete distribution:

$$P(y = 0) = \frac{\lambda^0 e^{-\lambda}}{0!} = e^{-\lambda}$$

$$P(y = 1) = \frac{\lambda^1 e^{-\lambda}}{1!} = \lambda e^{-\lambda}$$

$$P(y = 2) = \frac{\lambda^2 e^{-\lambda}}{2!} = \frac{1}{2}\lambda^2 e^{-\lambda}$$

……



  - $\sum_{k=0}^{+\infty} \frac{\lambda^k e^{-\lambda}}{k!} = 1$
  - $E(y) = \lambda\ (> 0), \qquad Var(y) = \lambda$

Poisson Regression Model

- The Poisson Regression Model:

$$E(y|X) = \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k) = \exp(X\beta)$$

  - $\exp(X\beta)$ is the same as $e^{X\beta}$
  - Notice that $E(y|x) = \exp(X\beta)$ is guaranteed to be always positive regardless of $X\beta$ value
  - y follows the Poisson distribution with $\lambda = \exp(X\beta)$

$$P(y = k|X) = \frac{\exp(X\beta)^k \, e^{-\exp(X\beta)}}{k!}, \qquad k = 0,1, \ldots$$

Estimating Poisson Model

- Likelihood function:

$$f(y_i|X_i) = \frac{\exp(X_i\beta)^{y_i}\, e^{-\exp(X_i\beta)}}{y_i!}$$

- Log-likelihood function for each observation i:

$$\ell_i(\beta) = \ln f(y_i|x_i) \;\sim\; y_i X_i\beta - \exp(X_i\beta)$$

- Log-likelihood function for the whole sample:

$$\mathcal{L}(\beta) = \sum_i \ell_i(\beta) \sim \sum_i y_i X_i\beta - \exp(X_i\beta)$$

- Tobit model can be estimated by **maximum likelihood estimation (MLE)**

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}}\, \mathcal{L}(\beta)$$

Interpreting the Coefficients

- β now has a direct interpretation in a **log-linear** model

$$\ln[E(y|X)] = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$$

$$\frac{\partial}{\partial x_j} \ln[E(y|X)] = \beta_j$$

$$\frac{\partial E(y|X)}{\partial x_j} \cdot \frac{1}{E(y|X)} = \beta_j$$

- β_j measures (approximately) the ***percentage change*** of the expected value of y when x_j changes by one unit
  - E.g., $\beta_j = 0.25$ means that if x_j increases by one unit, $E(y|X)$ will increase by roughly 25%

Interpreting the Coefficients (Cont'd)

- Partial (marginal) Effects

$$\frac{\partial E(y|X)}{\partial x_j} = \frac{\partial \exp(X\beta)}{\partial x_j} = \exp(X\beta) \cdot \beta_j$$

- **Average Partial Effect (APE)**: $\frac{1}{n}\sum_{i=1}^{n} \exp\left(X_i\hat{\beta}\right) \cdot \hat{\beta}_j = \left(\frac{1}{n}\sum_{i=1}^{n} \hat{y}_i\right) \cdot \hat{\beta}_j = \bar{y} \cdot \hat{\beta}_j$
  - $\hat{y}_i \equiv \exp\left(X_i\hat{\beta}\right)$
  - $\frac{1}{n}\sum_{i=1}^{n} \hat{y}_i = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y}$ is guaranteed by the MLE first order condition

Limitations of Poisson Model

- Limitations of Poisson Model:
  - $Var(y) = E(y)$
  - In reality, $Var(y) > E(y)$ ***Overdispersion***
    $Var(y) < E(y)$ ***Underdispersion*** (less common)

- Examining Potential Overdispersion
  - Formal tests
  - Estimate the dispersion parameter σ^2

$$Var(y|X) = \sigma^2 E(y|X)$$

    - $\sigma^2 > 1$ Overdispersion
    - $\sigma^2 < 1$ Underdispersion
  - A consistent estimate of σ^2:

$$\hat{\sigma}^2 = \frac{1}{n-k-1}\sum_{i=1}^{n}\frac{(y_i - \hat{y}_i)^2}{\hat{y}_i}$$

    - $\hat{y}_i = \exp(X_i\hat{\beta})$, k is the number of X variables
    - *$\hat{\sigma}^2$ notably greater than 1* indicating overdispersion

Summary

- Poisson distribution
- Poisson Regression model
  - Model setup
  - Estimation
- Interpret Poisson model results:
  - Direct interpretation of β
  - Usual partial effects
- Overdispersion
  - Estimate the overdispersion parameter

DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Count Data Models: Negative Binomial Models

Learning Objectives

- Develop Negative Binomial models
  - Adding random effects to a Poisson model
  - Gamma-Poisson mixture
  - Deriving the likelihood function
- Explain the relationship between Negative Binomial and Poisson models
  - Convergence in the extreme case
  - Overdispersion
- Use likelihood-based goodness-of-fit
  - AIC, BIC

Negative Binomial Model (1/4)

- Negative Binomial Model
  - An extension of the Poisson model to allow overdispersion
  - It can be derived as a Poisson model with ***random effects***

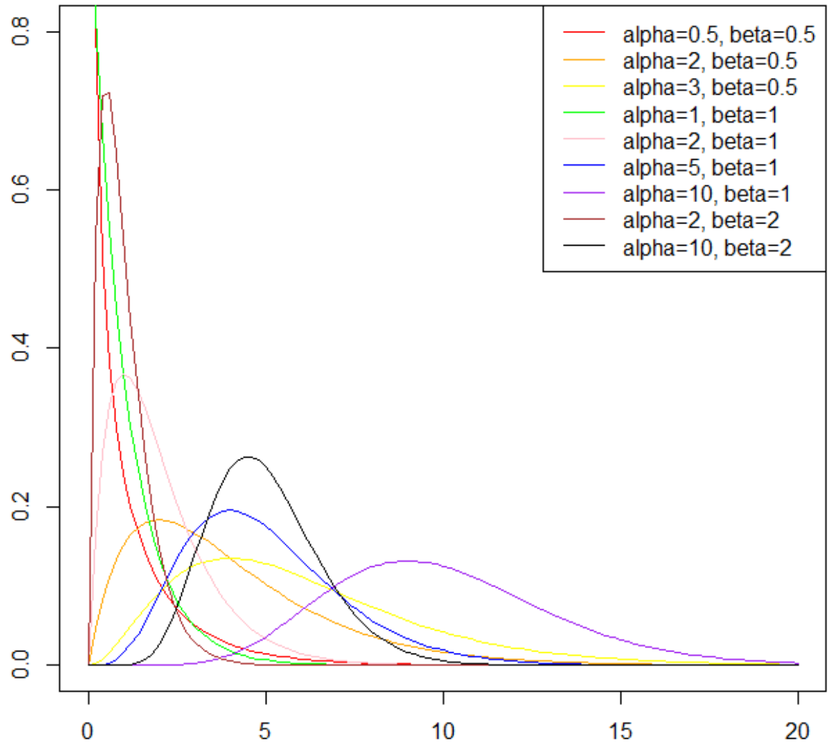
$$\boldsymbol{E(y_i|X_i) = v_i \cdot \mu_i = v_i \cdot \exp(X_i\beta)}$$

  - Here, v_i is the *individual random coefficient*, which is <u>*unobservable*</u> and allowed to <u>*vary*</u> from individual to individual
  - v_i is <u>*positive*</u> and assumed to follow ***Gamma distribution*** $Gamma(v_i; \alpha, \beta)$:

$$f(v; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} v^{\alpha-1} e^{-\beta v}$$

  - $E(v) = \frac{\alpha}{\beta} \quad Var(v) = \frac{\alpha}{\beta^2}$
  - Let $\alpha = \beta = \theta > 0$, so $E(v) = 1$, $Var(v) = \frac{1}{\theta}$

$$f(v; \theta) = \frac{\theta^\theta}{\Gamma(\theta)} v^{\theta-1} e^{-\theta v}$$



Negative Binomial Model (2/4)

- Substitute the Gamma distribution of the random coefficient into the Poisson distribution, and integrate out the random coefficient

$$P(y=k) = \int_0^{+\infty} \frac{(\upsilon\mu)^k e^{-\upsilon\mu}}{k!} \cdot \frac{\theta^\theta}{\Gamma(\theta)} \upsilon^{\theta-1} e^{-\theta\upsilon} d\upsilon$$

$$= \cdots$$

$$= \frac{\boldsymbol{\Gamma(k+\theta)}}{\boldsymbol{\Gamma(\theta)k!}} \left(\frac{\boldsymbol{\theta}}{\boldsymbol{\theta+\mu}}\right)^{\boldsymbol{\theta}} \left(\frac{\boldsymbol{\mu}}{\boldsymbol{\theta+\mu}}\right)^{\boldsymbol{k}}$$

$$\sim NegBinom(\theta, \mu)$$

- This is the Negative Binomial distribution with the *dispersion parameter* θ and *mean* μ
- Negative Binomial is the outcome of a ***Gamma-Poisson mixture*** model

Negative Binomial Model (3/4)

- Substitute in $\mu = \exp(X\beta)$, we have the Negative Binomial Model:

$$P(y_i = k) = \frac{\Gamma(k+\theta)}{\Gamma(\theta)k!}\left(\frac{\theta}{\theta + \exp(X_i\beta)}\right)^{\theta}\left(\frac{\exp(X_i\beta)}{\theta + \exp(X_i\beta)}\right)^{k}$$

$$\sim NegBinom(\theta, \exp(X_i\beta))$$

- Likelihood function:

$$\mathcal{L}(\beta,\theta) = \prod_i \frac{\Gamma(y_i+\theta)}{\Gamma(\theta)y_i!}\left(\frac{\theta}{\theta + \exp(X_i\beta)}\right)^{\theta}\left(\frac{\exp(X_i\beta)}{\theta + \exp(X_i\beta)}\right)^{y_i}$$

- The MLE estimator of (β, θ) is: $(\hat{\beta}, \hat{\theta}) = \underset{\beta,\theta}{\operatorname{argmax}}\, \mathcal{L}(\beta,\theta)$

- *Side note*: The name "negative binomial" comes from the fact that the probability distribution resembles the probability of the number of *failures* observed before a certain number of successes are reached in repeated Bernoulli trials.

Negative Binomial Model (4/4)

- Notice that **when $\theta \to +\infty$, the negative binomial distribution *converges* to the Poisson distribution**

$$\lim_{\theta \to +\infty} P(X = k) = \lim_{\theta \to +\infty} \frac{\mu^k}{k!} \cdot \frac{\Gamma(k+\theta)}{\Gamma(\theta)(\theta+\mu)^k} \cdot \frac{1}{\left(1+\frac{\mu}{\theta}\right)^\theta}$$

$$= \frac{\mu^k}{k!} \cdot 1 \cdot \frac{1}{e^\mu} \quad \sim Poisson(\mu)$$

- For negative binomial distribution, $\boldsymbol{E(X) = \mu}$, and

$$\boldsymbol{Var(X) = \mu + \frac{1}{\theta}\mu^2} > E(X)$$

- So negative binomial is always over-dispersed
- Small θ indicates *significant* overdispersion;
- Large θ indicates *little* overdispersion
  - $\theta \to +\infty$, $Var(X) \to E(X)$, no overdispersion, converges to Poisson

Poisson vs Negative Binomial

- Note that for both Poisson and Negative Binomial models,

$$E(y|X) = \mu = \exp(X\beta)$$

- Therefore, the Poisson estimator is still consistent *<u>even if the distributional assumption is incorrect</u>*
- Using Poisson MLE to estimate the model without assuming the Poisson distribution is entirely correct, this is called **Quasi-Maximum Likelihood Estimation (QMLE)**
- Even with overdispersion, Poisson model is still good enough <u>if we are only interested in the *expected value* of y</u>
- If we are interested in interpreting the *<u>distribution</u>* (e.g., estimating the probability of y taking certain values), the negative binomial model is helpful in properly accounting for overdispersion

Goodness of Fit: AIC, BIC

- Akaike information criterion (AIC)

$$AIC = 2k - 2\ln(\hat{L})$$

- Bayesian information criterion (BIC)

$$BIC = k\ln(n) - 2\ln(\hat{L})$$

- $\hat{L}$: the *maximized* likelihood
- k: number of parameters in a model; n: number of data observations
- Both measure the goodness-of-fit of models estimated by MLE
- Both are based on the *maximized* log-likelihood
- Both penalize the number of parameters introduced in the model
- **<u>The smaller, the better</u>**

Summary

- Develop Negative Binomial models
  - Adding random effects to a Poisson model
  - Gamma-Poisson mixture
  - Deriving the likelihood function
- Relationship between Negative Binomial and Poisson models
  - Convergence in the extreme case
  - Overdispersion
- Likelihood-based goodness-of-fit
  - AIC, BIC